Xixian Niu

ECARES and I3h, Université Libre de Bruxelles
xixian_niu@outlook.com · +32 456 16 58 92

EDUCATION

Ph.D. in Economics, Université Libre de Bruxelles (ECARES and I3h) expected 2027
Supervisors: Micael Castanheira and Mathias Dewatripont

Research Master in Economics, ECARES, Université Libre de Bruxelles 2023
Supervisor: Mathias Dewatripont

M.Sc. in Quantitative Economics (QEM), Paris 1 Panthéon-Sorbonne and UCLouvain 2022
Supervisor: Hervé Moulin

B.Sc. in Mathematical Statistics, School of Economics, Xiamen University 2018

RESEARCH FIELDS

Primary: Economics of Innovation, Health Economics, Industrial Organization
Secondary: Applied Microeconomics, Applied Econometrics

JOB MARKET PAPER

"Asymmetric Intellectual Property Rights and Follow-On Innovation: Evidence from Drug Repurposing"

Theories of cumulative innovation treat an upstream patent as a gate, assuming symmetric protection: follow-on innovators can protect their own inventions from imitation. I show that when they cannot, the upstream patent also acts as a shield, because imitating a licensed follow-on infringes it, so longer upstream protection can increase follow-on innovation. Drug repurposing exhibits this asymmetry: a molecule's patent covers every use, but new-use rights are narrow and hard to enforce. Using FDA-approved new molecular entities, 1985–2024, and instrumenting effective exclusivity with FDA review time, I estimate that one more year of exclusivity raises expected post-approval indications by about 34 percent, with the marginal effect vanishing beyond roughly 15 years. Originators and entrants both respond positively, as the shield predicts. The results point to a failure to protect new uses, not a case for longer patents.

PUBLICATIONS

S. Mermelstein, A. Gentilini, C. Yue, **X. Niu**, M. Shaikh, M. Dewatripont, H. Stevens, and M. Miraldo. "Do Policy Incentives Align with Pharmaceutical Innovation Challenges? A Retrospective Analysis of 42 Countries, 2000–2022." *Health Economics*, 2027 (forthcoming).

S. Akodad, **X. Niu**, B. Secades, and H. Stevens. "Impact of Drug Repurposing between 1985 and 2024 on Pharmaceutical Innovation." *Communications Medicine*, 6:84, 2026. doi.org/10.1038/s43856-025-01344-1

WORKING PAPERS

"Unbundling Global Innovation Policies: Mechanism Effects, Spillovers, and the Equitable Allocation of Pharmaceutical R&D." With C. Yue, S. Mermelstein, M. Miraldo, H. Stevens, M. Shaikh, and M. Dewatripont.

Innovation policies bundle several incentive mechanisms with offsetting effects, so estimated policy-level effects identify no single margin. Decomposing 1,100 policy introductions into six mechanisms and linking them to 1.9 million R&D grants across 42 countries (2000–2021), we exploit within-policy variation in mechanism composition to separate own-country effects, cross-country spillovers, and allocation relative to disease burden. Revenue subsidies raise R&D but redirect it toward commercially lucrative diseases and high-income markets, widening the gap between private and social returns. Mechanisms that lower regulatory and information frictions yield larger and more persistent gains, concentrated where profit incentives are weakest, and generate positive spillovers across borders. Mandated IP-sharing lowers appropriability and depresses short-run entry. Success-contingent rewards dominate unconditional transfers, consistent with reduced moral hazard and better selection.

WORK IN PROGRESS

"Patent Box Tax Relief and Orphan Drug Development."

Nine EU countries have extended preferential IP-box tax rates to orphan-drug income since 2015. Using difference-in-

differences on a firm–country–year panel of 78,060 observations across 30 EMA member states (2000–2024), I find that adoption raises orphan-drug launches by roughly 75 percent. The effect is confined to large firms, with no response from SMEs and no increase in designations or registrations, indicating that the policy stimulates commercialization rather than innovation.

RESEARCH PROJECTS

Pharmaceutical innovation policy in 42 countries, Core Contributor — Brussels · 2023 – 2026
Hi-Prix Work Package 7, co-led by ULB and Imperial College London (Horizon Europe, €5.3M, 18 partner organizations in 10 countries); doctoral researcher funded by the project
Output: "Incentives for Pharmaceutical Innovation and Equitable Access to Innovation." Policy report for Hi-Prix, 2026. With M. Miraldo, H. Stevens, C. Yue, S. Mermelstein, A. Gentilini, M. Shaikh, and M. Dewatripont.

RESEARCH EXPERIENCE

Visiting Researcher, Imperial College London — May 2023 – May 2024
Centre for Health Economics and Policy Innovation, host: Prof. Marisa Miraldo. Empirical research on pharmaceutical innovation policy within Hi-Prix; co-authored the two papers and the policy report listed above.

Research Intern, Fair Division project, Paris 1 Panthéon-Sorbonne — 2021 – 2022
Supervisor: Prof. Hervé Moulin

PRESENTATIONS

2026 European Winter Meeting of the Econometric Society (EWMES), Warsaw (forthcoming)
2025 Hi-Prix Consortium Annual Conference, Brussels (poster)
2024 Hi-Prix Consortium Annual Conference, Lisbon

FELLOWSHIPS AND AWARDS

Ph.D. Fellowship, Horizon Europe Hi-Prix project — 2023 – 2026

SKILLS AND LANGUAGES

Programming: Python, R, Stata, Quarto / R Markdown, LaTeX (proficient); SQL, Git (basics).
Data engineering: API-based collection (Dimensions, Cortellis); web scraping (Requests, BeautifulSoup, Selenium); PDF text extraction; LLM-assisted extraction from unstructured documents; record linkage and server-based workflows for datasets with millions of records.
Methods: Panel data, applied econometrics and causal inference, including difference-in-differences, instrumental variables (including shift-share), regression discontinuity, and synthetic control; industrial-organization modeling; machine learning for feature selection (LASSO, random forest); questionnaire design and expert interviews.
Domain data: FDA Orange Book and Patent Term Extension records; EMA orphan-drug designations; Cortellis Competitive Intelligence; Dimensions (publications, patents, grants, clinical trials); ICD-11; cross-country R&D tax, IP-box and HTA policy.
Languages: Chinese (native), English (fluent).

REFERENCES

Micael Castanheira, FNRS Research Director and Professor, ECARES, ULB — micael.castanheira@ulb.be
Mathias Dewatripont, Professor of Economics, ECARES, ULB — mathias.dewatripont@ulb.be
Mattia Nardotto, Associate Professor of Economics, ECARES, ULB — mattia.nardotto@ulb.be

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